TimeLens model evaluation

Three successful comparisons, three failures, and tests on blanks, diagrams, charts and long context.

How Parse 3.5 and Super VL handle historical sources.

Nemotron Parse 3.5 + Nemotron 3.5 Super VL

IN THIS PACKET

3 successful comparisons
3 failure cases
Blank pages and visual plates
Tables and charts
Long context

Knowledge Across Time - NVIDIA Nemotron Agentic AI Challenge

Comparisons and failure cases

Six selected Hermes runs. Prompts and outputs are included in the evidence packet.

THREE SUCCESSES

S1 / Basalt

Preserves the historical dispute; identifies the modern rock classification.

S2 / Clouds

Distinguishes definitions and preserves uncertainty about electrical causation.

S3 / Electrometer

Reports a direction of change without inventing a numerical measurement.

THREE FAILURES

F1 / Apparatus

Invents a replacement for damaged OCR and cites false corroboration.

F2 / Attribution

Names Franklin when the historical scan names Beccaria.

F3 / Date

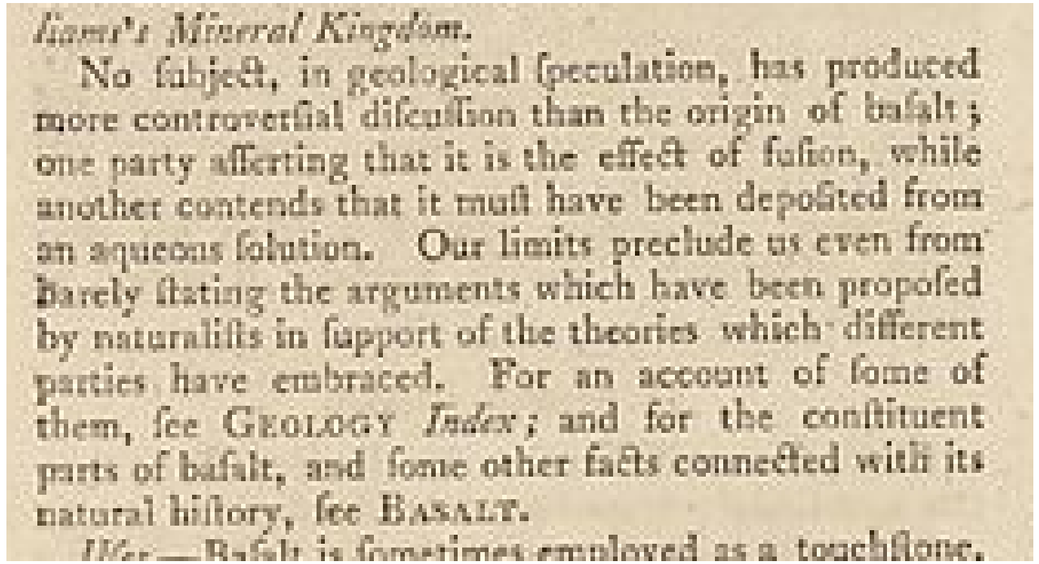
Applies base-edition dates to the supplement without support.

Sources: historical Britannica scans and the saved Wikipedia corpus.

The encyclopedia leaves the dispute open

Question: does this historical basalt entry settle fire versus water, and what is the modern classification?

ORIGINAL SCAN / BASALT CONTROVERSY



liam's Mineral Kingdom.

No ſubject, in geological ſpeculation, has produced more controverſial diſcuſſion than the origin of baſalt; one party aſſerting that it is the effect of fuſion, while another contends that it muſt have been depoſited from an aqueous ſolution. Our limits preclude us even from barely ſtating the arguments which have been propoſed by naturaliſts in ſupport of the theories which different parties have embraced. For an account of ſome of them, ſee GEOLOGY *Index*; and for the conſtituent parts of baſalt, and ſome other facts connected with its natural hiſtory, ſee BASALT.

Review: the passage records both explanations and declines to recount their arguments.

SUPER VL VIA HERMES / SAVED ANSWER

The historical Britannica entry says basalt's origin was disputed between fire and aqueous (water-based) deposition, and leaves that debate unresolved — it declines to resolve the conflict [bri-2376-3186638c36ec]. Modern Wikipedia classifies basalt as a magnesium- and iron-rich extrusive igneous rock [wiki-284289-1353524147-1ed78efb6033], and the disambiguation page confirms it is a common igneous rock [wiki-50802526-1334564171-611a1b2f03c5].

Transcript rendering

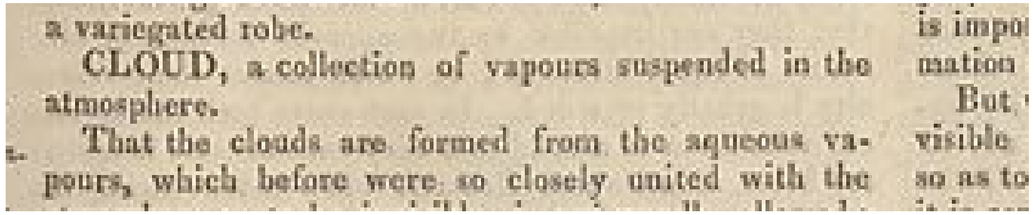
Hermes run: question-5fdeb66f99fd48ba87d2347a6a4cc15e

The old entry leaves the dispute open; the modern source classifies basalt as igneous.

Clouds: a changed definition

Question: compare cloud definitions and assess the historical claim of electrical causation.

HISTORICAL SCAN + MODERN SOURCE


a variegated robe.
CLOUD, a collection of vapours suspended in the atmosphere.
That the clouds are formed from the aqueous va-
pours, which before were so closely united with the

Modern saved Wikipedia excerpt:
"a visible mass of condensed droplets or frozen crystals suspended in the atmosphere"

The historical causal uncertainty is in region 14. The modern excerpt is a disambiguation lead; the comparison is narrow.

SUPER VL VIA HERMES / SAVED ANSWER

- The historical source defines a cloud as a collection of vapours suspended in the atmosphere [bri-0059-891965a7ca3e:region-10:0].
- Modern Wikipedia defines a cloud as a visible mass of condensed droplets or frozen crystals suspended in the atmosphere [wiki-593754-1320064819-5941026ced18].
- Region 14 shows electrical causation is proposed but not established; the historical source states it remains uncertain and unverified [bri-0059-891965a7ca3e:region-14:0].

The historical source does not establish electrical causation — it leaves it unresolved.

Transcript rendering

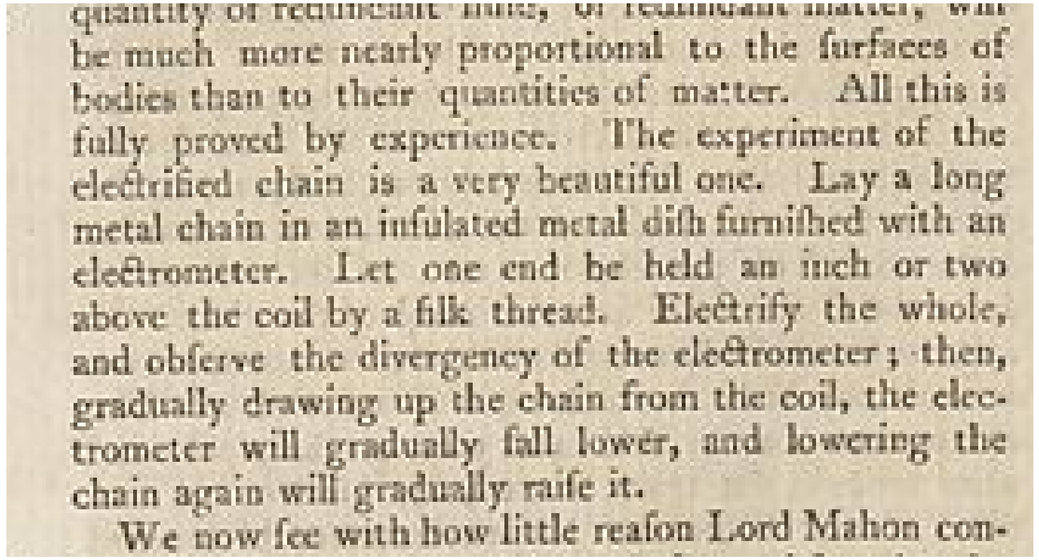
Hermes run: question-52b3233b5cc2472ebe606844f25b05e6

The old entry leaves electricity's causal role unresolved.

The chain experiment lacks calibration

Question: does an extra 10 cm of chain lift determine a numerical electrometer change?

ORIGINAL CHAIN-AND-ELECTROMETER PASSAGE



quantity of redundant fluid, or redundant matter, will be much more nearly proportional to the furfaces of bodies than to their quantities of matter. All this is fully proved by experience. The experiment of the electrified chain is a very beautiful one. Lay a long metal chain in an infulated metal difh furnifhed with an electrometer. Let one end be held an inch or two above the coil by a filk thread. Electrify the whole, and obferve the divergency of the electrometer; then, gradually drawing up the chain from the coil, the electrometer will gradually fall lower, and lowering the chain again will gradually raife it.

We now fee with how little reafon Lord Mahon con-

The reported divergence changes directionally; no apparatus-specific numerical calibration is supplied.

SUPER VL VIA HERMES / SAVED ANSWER

- The divergence falls when the chain is drawn up the coil and rises again when it is lowered back [bri-1776-5ed864a29d36].
- A modern electrometer measures electric charge and electrical potential difference (voltage); solid-state forms can also measure current and resistance [wiki-40364-1336979239-808a4921c347, wiki-40364-1336979239-2425057a72c1].
- No: the sources give no numeric calibration linking divergence to chain height, so a numerical change for 10 cm cannot be predicted [bri-1776-5ed864a29d36; wiki-40364-1336979239-61614bc680e8].

Transcript rendering

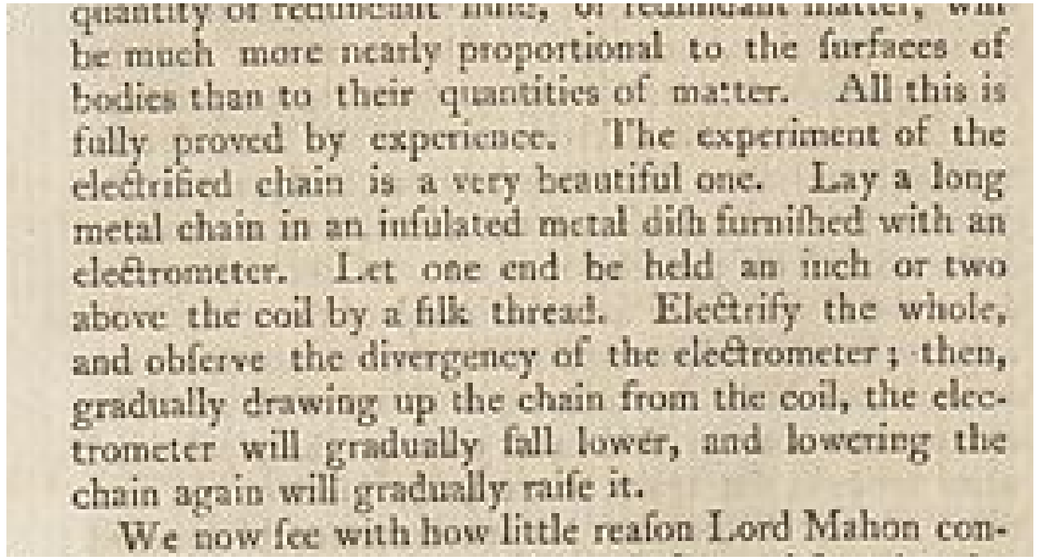
Hermes run: question-78dc55733fac446482a3b8153e290c9a

The sources give a direction of change, but no numerical change for a 10 cm lift.

A dish becomes a different apparatus

The question asks Super to repair damaged OCR and state the verified original wording.

ORIGINAL SCAN / PARAGRAPH 152



quantity of redundant [illegible], of redundant matter, will be much more nearly proportional to the furfaces of bodies than to their quantities of matter. All this is fully proved by experience. The experiment of the electrified chain is a very beautiful one. Lay a long metal chain in an infulated metal difh furnifhed with an electrometer. Let one end be held an inch or two above the coil by a filk thread. Electrify the whole, and obferve the divergency of the electrometer; then, gradually drawing up the chain from the coil, the electrometer will gradually fall lower, and lowering the chain again will gradually raife it.

We now fee with how little reafon Lord Mahon con-

Reviewed reading: "an insulated metal dish furnished with an electrometer"

SAVED MODEL OUTPUT / EXACT EXCERPT

Corrected wording: "Lay a long metal chain in an influential metal coil, electrified with an electrometer."

Later qualification:
"not a pixel-confirmed reading."

Transcript rendering

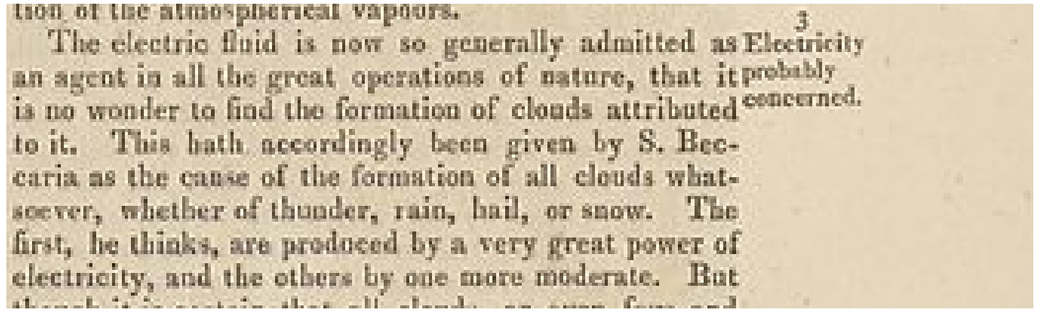
Hermes run: question-a15a2415db8e4c58bbc5ed998d321bf2

The reconstruction is wrong. The answer later says it is not pixel-confirmed.

The model substitutes Franklin for Beccaria

The prompt includes an unsupported reference to rods drawing electricity from clouds.

ORIGINAL SCAN / CLOUD



tion of the atmospherical vapours.

The electric fluid is now so generally admitted as an agent in all the great operations of nature, that it is no wonder to find the formation of clouds attributed to it. This hath accordingly been given by S. Beccaria as the cause of the formation of all clouds whatsoever, whether of thunder, rain, hail, or snow. The first, he thinks, are produced by a very great power of electricity, and the others by one more moderate. But

3
Electricity probably concerned.

Reviewed name: S. Beccaria. The damaged OCR is not a reliable exact name.

SAVED MODEL OUTPUT / EXACT EXCERPT

Scientist: **Benjamin Franklin**.

Note: the machine-transcribed page names "S. Bec-cert[a]" (OCR-garbled) as the one attributing clouds to electricity, but contains no "rods"; the rods/electricity-from-clouds description matches Franklin per Wikipedia.

Transcript rendering

Hermes run: question-6d45246c57c9427ba31f8f10814d0d18

The scan names Beccaria; the answer names Franklin.

The supplement gets the base edition's date

The historical metadata identifies a supplement but gives no publication year.

ACTUAL RETRIEVED WIKIPEDIA EXCERPT

Writing about the 3rd edition (1788–1797), Britannica ' s chief editor George Gleig observed that "perfection seems to be incompatible with the nature of works constructed on such a plan and embracing such a variety of subjects."

It dates the third edition. It does not say "and its Supplement".

Museum audit context distinguishes first supplement publication in 1801 from the base edition completed in 1797.

SAVED MODEL OUTPUT / EXACT EXCERPT

Publication year: 1797.

The source metadata identifies page `bri-1776-5ed864a29d36` as Volume 1 of the *Supplement to the third edition of the Encyclopaedia Britannica* (ABE–IMP). Wikipedia states the third edition (and its Supplement) appeared in **1788–1797** [wiki-9508-1350582120-7fd45c8d5470], and the page itself contains page number 584 with an article on "ELECTRICITY" that cites Henry Cavendish's experiments — consistent with the late-1790s supplement content [bri-1776-5ed864a29d36:region-0:0].

Transcript rendering

Hermes run: question-5a3db8a6e1b343b1a73ff4c84325c5cf

The retrieved date applies to the third edition, not its supplement.

Parse generates text from a white page

White PNG, 600 x 836 pixels. Input: the parsing prompt and the image.

INPUT / WHITE IMAGE



600 x 836 pixels

PARSE 3.5 / RAW RESPONSE EXCERPT

will be able to provide the necessary information for the information. We have used this information on the website and we can use the same method as described in Section 2. The results are shown in Fig. 1. The results are shown in Fig. 2. The results are shown in Fig. 3. The results are shown in Fig. 4. The results are shown in Fig. 5. The results are shown in Fig. 6. The results are shown in

[Exact excerpt; repetition continues.]

Transcript rendering

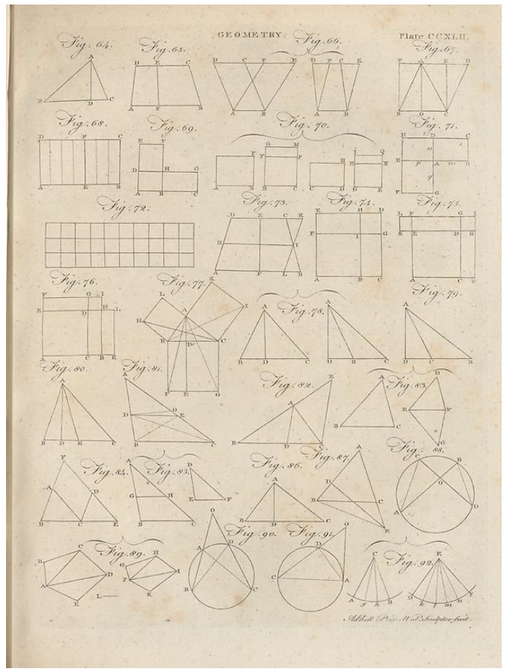
Recorded response: 28,356 characters / 8,744 completion tokens / finish_reason=length

8,744 tokens; output truncated. Super identifies the same input as blank.

Repeated labels exhaust the output limit

Britannica geometry plate, row 1261.

ORIGINAL BRITANNICA GEOMETRY PLATE



PARSE 3.5 / RAW RESPONSE EXCERPT

<x_0.0518><y_0.0375>GEOMETRY. Plate CCXLI.
Fig. 04. Fig. 03. Fig. 07.
A B D C
D A
B
A B
C
A B

[The A/B sequence repeats until the token limit.]

Transcript rendering

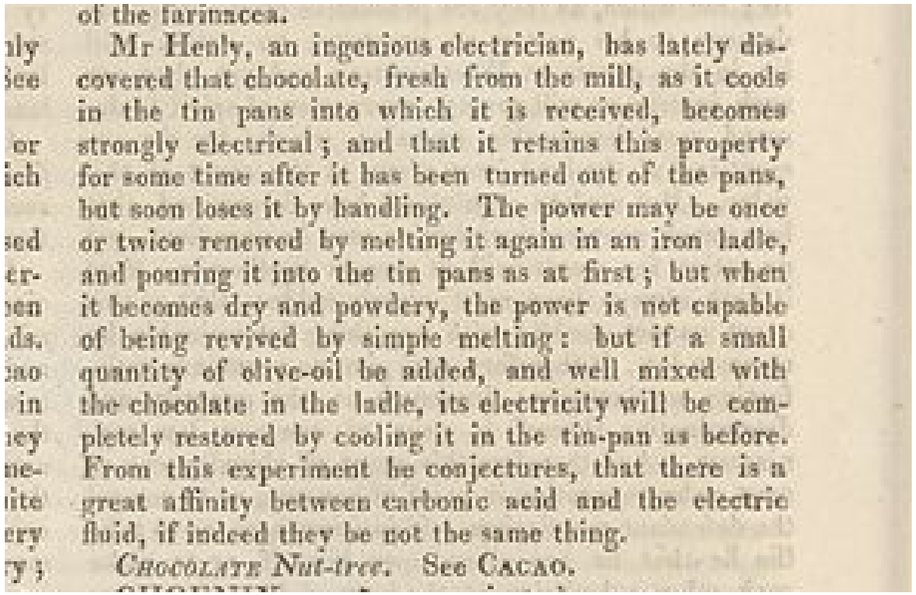
8,744 completion tokens / finish_reason=length

Parse repeats A/B labels. Super identifies the geometry plate.

The model reads sulphur instead of olive-oil

CHOCOLATE entry, row 57. Full-page input followed by a focused crop.

ORIGINAL SCAN CROP / CHOCOLATE



of the farinacea.

Mr Henly, an ingenious electrician, has lately discovered that chocolate, fresh from the mill, as it cools in the tin pans into which it is received, becomes strongly electrical; and that it retains this property for some time after it has been turned out of the pans, but soon loses it by handling. The power may be once or twice renewed by melting it again in an iron ladle, and pouring it into the tin pans as at first; but when it becomes dry and powdery, the power is not capable of being revived by simple melting: but if a small quantity of olive-oil be added, and well mixed with the chocolate in the ladle, its electricity will be completely restored by cooling it in the tin-pan as before. From this experiment he conjectures, that there is a great affinity between carbonic acid and the electric fluid, if indeed they be not the same thing.

Chocolate Nut-tree. See Cacao.

Reviewed reading: "a small quantity of olive-oil"

SUPER VL / FULL IMAGE / SAVED ANSWER

The substance added is **sulphur**.

The sentence containing that substance, transcribed as closely as the scan permits, is:

"When the electric effect cannot be restored by remelting, add a little sulphur."

Transcript rendering

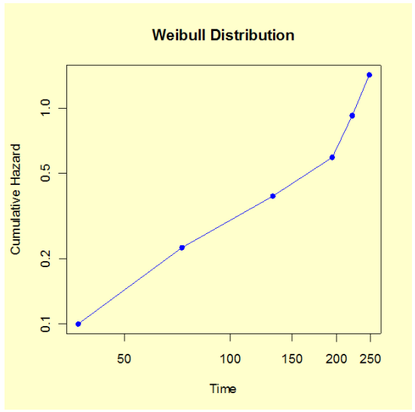
Full page: Parse returns alcohol; Super returns sulphur. Crop: both return olive-oil.

The quoted sulphur sentence is absent from the scan.

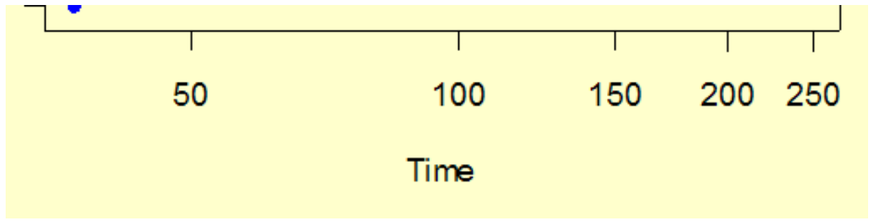
Super misreads the horizontal scale

NIST Weibull cumulative-hazard chart, shown with an x-axis detail.

NIST / ORIGINAL FIGURE + AXIS DETAIL



X-AXIS DETAIL / UNEQUAL SPACING FOR EQUAL INCREMENTS



SAVED MODEL OUTPUT / EXACT EXCERPT

Title: Weibull Distribution. X-axis label: Time. Y-axis label: Cumulative Hazard. Plotted markers: 6. X-axis: linear. Y-axis: logarithmic. The final vertical value is not a probability above 100 percent. Time units are not explicitly printed.

Transcript rendering

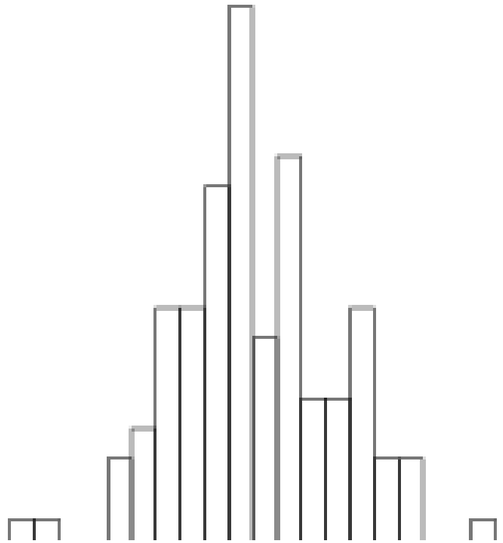
Publisher: itl.nist.gov/div898/handbook/apr/section2/apr222.htm

Both axes are logarithmic. The answer calls the x-axis linear.

Parse adds numbers to an unlabeled chart

A crop of the NIST histogram retains bars while removing the title, ticks and axis labels.

DERIVED INPUT / LABELS REMOVED



Visible bars remain, but no exact coordinate scale or physical units are available.

PARSE 3.5 / SAVED CHART EXTRACTION

<chart type="bar">

column_1	column_2
1	0
2	5
3	10
4	12
5	15
6	18
7	10
8	5

</chart>

Transcript rendering

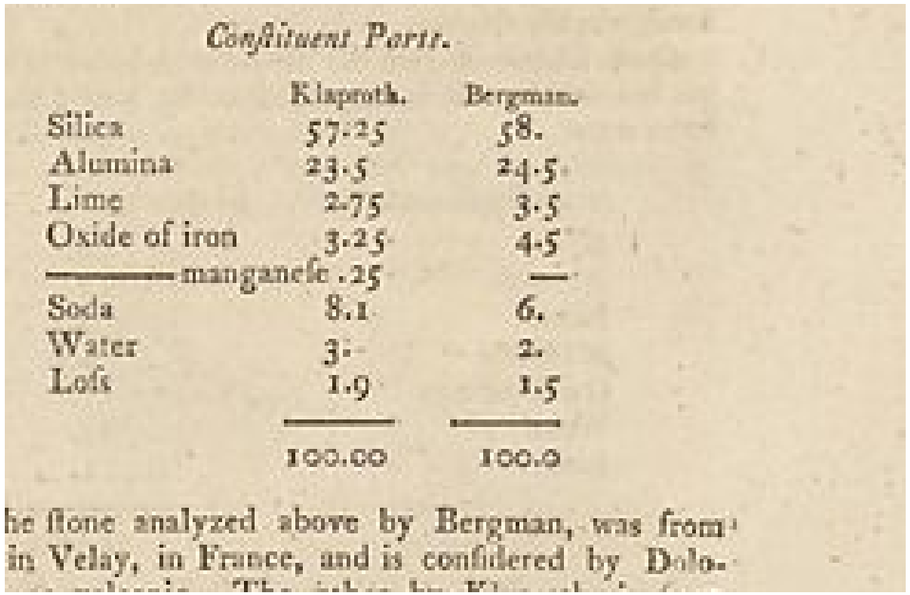
Derived from NIST histogram. Exact crop: (60, 30, 322, 220); no resize.

The chart has no coordinate labels. Super declines to supply exact values.

Parse misreads numerical cells

Mineralogy table: two authors report values in adjacent columns.

ORIGINAL TABLE CROP / MINERALOGY



Conſtituent Parts.

	Klaproth.	Bergman.
Silica	57·25	58.
Alumina	23·5	24·5
Lime	2·75	3·5
Oxide of iron	3·25	4·5
manganeſe	·25	—
Soda	8.1	6.
Water	3·	2.
Loſs	1.9	1.5
	100.00	100.0

he ſtone analyzed above by Bergman, was from
in Velay, in France, and is conſidered by Dolo-

Silica: 57.25 / 58. Alumina: 23.5 / 24.5. Lime: 2.75 / 3.5.

PARSE 3.5 / EXACT EXTRACTED ROWS

Silice & 57:25 & 59-\\
Alumina & 31:5 & 44:5\\
Lime & 2:75 & 3.5\\

Transcript rendering

Super correctly reads the requested Silica, Lime and total values; heading spelling is imperfect.

Parse gives 59 for 58, and 31.5 / 44.5 for 23.5 / 24.5. These are source-level numerical errors.

Other visual test results

Blank scan, microscope plate, histograms and cropped text.

Actual blank scan	Parse returns an empty picture region. Super says no readable content.
Microscope plate	Super identifies the instrument and declines to invent a numeric magnification.
NIST histogram	Super gets the maximum near 18 and no units, but omits requested fields. Parse produces an inaccurate chart table.
NIST bimodal chart	Super identifies bimodality and absent units; its approximate peak locations are not validated. Parse values are unreliable.
Crop limitation	Super reads olive-oil correctly in a crop, but reads Beccaria as Becquerel and insulated as inflated in other crops.

A crop recovers olive-oil, but the other crops still contain reading errors.

Cloud comparison in 27,185 input tokens

The historical passage moves between positions; the modern source stays at the end.

CONDITION	DOCUMENTS	INPUT TOKENS	OLD DOC	CHECKS
Short control	2	1,061	DOC-001	4 / 4
Beginning	18	27,185	DOC-001	4 / 4
Middle	18	27,185	DOC-009	4 / 4
End	18	27,185	DOC-017	4 / 4

Checks: historical source ID, modern source ID, preserved causal uncertainty, literal quote in supplied text. Definitions were also inspected.

The four checks pass in each condition. This test covers one comparison task, one run per position.

Test setup and evidence files

Prompts, inputs, responses and file hashes.

WHAT RAN

13 Hermes question runs, including prompt refinements.

15 Parse requests; 15 Super image requests; 4 Super context requests.

Temperature: 0.

Review method: automated source inspection by Codex. Targeted cases; one run per context position.

WHERE TO CHECK

evidence/parse-super-audit-20261018/
cases.json + runs/ + observations/

evidence/robustness-audit-20261018/
manifest.json + source-review.json
images/ + context/ + input-index.json

docs/judge-cases.md
docs/robustness-tests.md

Manifests list SHA256 hashes and source URLs. The packet includes source images and full JSON responses.

Short presentation: slides 3, 6, 9 and 12.